Linear Algebra and Matrix Analysis

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Collections from Linear Algebra Done Right, Strange's Lectures.

Horn and Rogers' Matrix Analysis, the first three chapters are standard materials. Then there are theories on

1. Hermitian Matrices

2. Matrix Norm Pinite and matrices with positive elements.

Then we have Matrix Calculus book. We may delve into Random Matrix Theory later.

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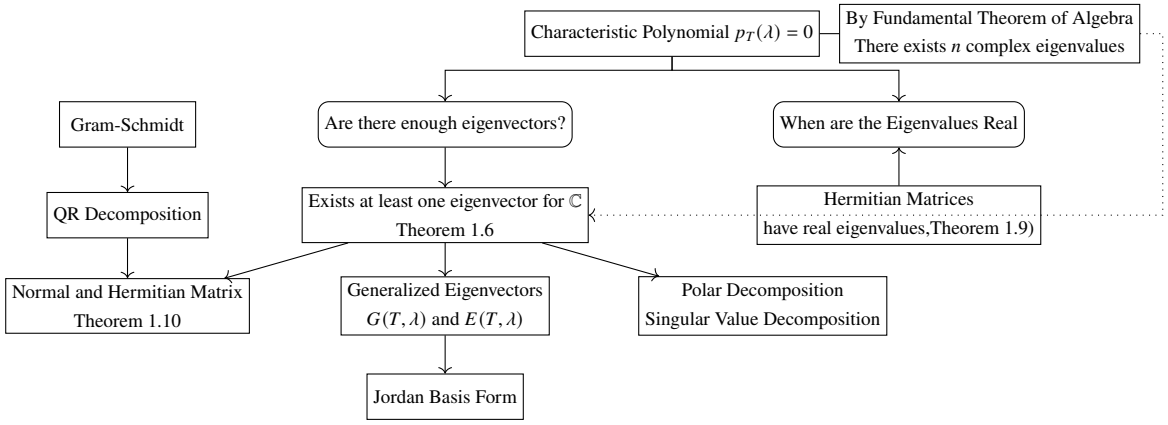
Chapter 1

Linear Algebra



$$\begin{tikzcd}
 & f_n \rightrightarrows f \arrow[dr] & f_n \rightrightarrows f \\
 & f_n \rightrightarrows f \text{ on } N^c \arrow[dr] & f_n \rightrightarrows f \text{ on } N^c \\
 & f_n \rightrightarrows f \text{ almost uniformly} \arrow[ddr] & f_n \rightrightarrows f \text{ almost uniformly} \\
 & f_n \to f \arrow[ddr] & f_n \to f \\
\text{Borel-Cantelli} & f_n \to f \text{ a.e.} & f_n \to f \text{ a.e.} \\
 & f_n \xrightarrow{\mu} f & f_n \xrightarrow{\mu} f \\
 & f_n \xrightarrow{L^1} f & f_n \xrightarrow{L^1} f \\
 & f_n \xrightarrow{L^p} f & f_n \xrightarrow{L^p} f \\
 & \mu_n \rightsquigarrow \mu & \mu_n \rightsquigarrow \mu
\end{tikzcd}$$

1.1 Vector Space and Inner Product Space

Vector Space. Basis, linear independent, subspace, rank and dimension, linear transformation, kernel and range. Direct sum: $U = V_1 \oplus V_2$, if $U = V_1 + V_2$ and $\bigcap V_j = \varnothing$.

1.1 Proposition (Isomorphism Between Linear Transformation and Matrices). *There is one-to-one linear transformation between $\mathcal{L}(X, Y)$ and $\mathbb{M}_{m\times n}$ for any choice of bases in X, Y.*

Proof. Simply consider the application of T on the basis. We will write $TE = FM$ for E, F being the bases and M being the matrix. □

Inner Product Space. Let $\langle\cdot|\cdot\rangle$ be mapping into the underlying field. Importantly $\langle x|y\rangle = \overline{\langle y|x\rangle}$.

$$x \perp y \iff \langle x|y\rangle = 0.$$

1.2 Cauchy-Schwarz. $\langle x|y\rangle \leq \|x\|\|y\|$ ◊

Proof. For any x, y we can define $x_{\|} = \frac{\langle x|y\rangle}{\|y\|^2} y$ and $x_{\perp} = x - x_{\|}$. Then $x_{\perp} \perp y$ and so by $\|x\|^2 = \|x_{\perp}\|^2 + \left\|x_{\|}\right\|^2$ and $\|x_{\perp}\| \geq 0$ we have the desired result. □

1.3 Riesz Representation. *For any linear functional $T \in \mathcal{L}(X, \mathbb{F})$ where X is an inner product spaces over $\mathbb{F}$, we have a unique x^* such that $T(x) = \langle x|x^*\rangle$* ◊

Proof. Consider a basis $E = [e_1, \ldots, e_n]$, $T(x) = T(E)x$, let $x^* = E\left[Te_1, \ldots, Te_n\right]^*$. The latter $*$ is complex conjugate. □

1.4 Gram-Schmidt. *Any basis E of an inner product space can be made into an orthonormal basis, o.n.b, by an upper-triangular trasformation G.* ◊

Proof. Let $G = \left[G_1, \ldots, G_n\right]$, then $G_1 = \left[\|e_1\|^{-1}, 0, 0, \ldots, 0\right]'$, $G_2 E = e_{2\perp 1}/\|e_{2\perp 1}\|$ and so on. □

Adjoint of T. By Riesz theorem, we have for any $T : \mathcal{L}(X, Y)$ over F, and fix $y \in Y$, $\langle Tx|y\rangle \in \mathcal{L}(X, \mathbb{F})$. Hence there exists unique x_y such that $\langle Tx|y\rangle = \left\langle x\middle|x_y\right\rangle$. Let the mapping $y \to x_y$ be the **adjoint** of T. It's in $\mathcal{L}(Y, X)$.

Self-Adjoint Operator. $T = T^*$.

1.2 Invariant Subspace

Invariant Subspace. If $Tx \in V \sqsubset U$ where $\sqsubset$ means linear subspace for all $x \in V$, then we say V is an **invariant subspace** of T. An **eigenspace** is $E(T, \lambda) = N(T - \lambda I)$. On eigenspace, T is simple, it's only scaling the vectors.

Remark. If we can decompose U into the direct sum of invariant subspaces, or better eigenspaces, then we can usually find basis that simplifies the representation of T.

1.5 Diagonalizable Linear Operator. *For $T \in \mathcal{L}(X)$, T is diagonalizable if for some E, and diagonal matrix Λ, $T(E) = E\Lambda$, which is equivalent to $X = \oplus_{j=1}^{m} E(T, \lambda_j)$, where there are m distinct eigenvalues.* ◊

Remark. We always have $\sum \dim E(T, \lambda_j) \leq \dim V$, when it's strictly less, we don't have eigen-decomposition.

1.6 Existence of Eigenvalue. *For any $T \in \mathcal{L}(X)$, there exist at least one **complex** eigenvalue-eigenvector pair.* ◊

Proof. For any linear operator, we can define polynomial of T. For any non-zero vector x,

$$x, Tx, \ldots, T^k x \tag{1.1}$$

is not linear independent for some $k \leq n$ (because $\dim V = n$), hence there exists $c_0 \neq 0, \ldots, c_k$ such that

$$\left(c_0 T^k + c_1 T^{k-1} + \cdots + c_k T^0\right) x = 0 \tag{1.2}$$

by Fundamental Theorem of Algebra, we have

$$(T - \lambda_1 I)(T - \lambda_2 I) \ldots (T - \lambda_k I) x = 0 \tag{1.3}$$

so for some λ_j, $N(T - \lambda_j I) \neq \{0\}$. □

Because $\lambda \in \sigma(T) \iff N(T - \lambda I) \neq 0 \iff \det\{T - \lambda I\} = 0$, by the Fundamental Theorem of Algebra, since that means $\dim R(T) < \dim X$.

1.7 Fundamental Theorem of Algebra. *For $T \in \mathcal{L}(X, Y)$, $\dim X = \dim N(T) + \dim R(T)$.* ◊

It's useful to analyse $\det\{\lambda I - T\} =: p_T(\lambda)$ which is called the **characteristic polynomial**. It's of order n. So by fundamental theorem of algebra, there exists n complex roots.

1.8 The Roots of $p_T(z)$ are Eigenvalues. *Let $z_1, \ldots, z_n$ be the roots of $p_T(z)$, then they are eigenvalues of T. Eigenvalues λ of T satisfies $p_T(\lambda) = 0$.* ◊

Now we have n complex eigenvalues. The next questions

1.9 Real Eigenvalues. *Let T be a Hermitian or self-adjoint operator, then every eigenvalue of T is real.* ◊

Proof. Let λ be an eigenvalue of T, let v be an associated eigenvector. Then let $B = T - \lambda I$, which is also self-adjoint,

$$0 = \langle Bv|v\rangle = \langle v|B^* v\rangle = \langle v|Bv\rangle = \langle v|Tv\rangle - \langle v|\lambda v\rangle = (\lambda - \bar{\lambda})\|v\|^2 \tag{1.4}$$

So $\lambda = \bar{\lambda}$. □

1.10 Normal Spectral Theorem. *Let T be a normal operator with underlying field $\mathbb{C}$, then T is (unitarily) diagonalizable.* ◊

Proof. For an o.n.b Q, $TQ = QR$, let R^H be the conjugate transpose of R, it's asscociated with T^*. We know R is upper triangular.

$$RR^H = R^H R \tag{1.5}$$

would imply that R is also diagonal. □

1.11 Spectral Theorem for Hermitian Matrices. ◊

1.3 Singular Value Decomposition

Chapter 2

Matrix Analysis

2.1 Chapter 1

2.1.1 Spectrum

2.1 **$\sigma(A)$ and $\sigma(p(A))$.** ◊

Corollary 2.1.1. $\sigma(A) \neq \varnothing$.

2.1.2 Details about Characteristic Polynomical

$$p_A(t) = \det\{tI - A\} = (t - \lambda_1)^{i_1} \dots (t - \lambda_k)^{i_k} \tag{2.1}$$

Hence the relationship between $E_k(A)$: sum of k-th principal minors and $S_k = \sum_i \prod_{j=1}^{k} \lambda_{i_j}$ and the coefficient in

$$p_A(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_0 \tag{2.2}$$

$$a_{n-k} = (-1)^k k! E_k(A) \quad E_k(A) = S_k(A) \tag{2.3}$$

2.1.3 Diagonalizability

2.2 **Theorem.** *There exists k independent eigenvectors* $\iff A \sim \begin{bmatrix} \Lambda_k & C \\ 0 & D \end{bmatrix}$. ◊

2.1.4 Commuting Family of Matrices

Let $\mathcal{F}$ be c.f.m, i.e., for any $A, B \in \mathcal{F}$, $AB = BA$. Then we have

2.3 Theorem. *$\mathcal{F}$ share at least one eigenvector.* ◊

Proof. Consider $\mathcal{F}$-invariant subspace W. Fix any A, there exists an eigenvector x of A, $x \in W$, $Ax = \lambda x$. Consider the subspace $W_{A,\lambda} = \{y \in W : Ay = \lambda y\}$, then $W_{A,\lambda}$ is $\mathcal{F}$-invariant. If we require W to have smallest dimension, then $W_{A,\lambda} = W$ and so $W = \bigcap_i W_{A_i,\lambda_i}$. Any nonzero $x \in X$ is an eigenvector for all $A_i \in \mathcal{F}$. □

2.4 Theorem. *If every $A \in \mathcal{F}$ is diagonalizable, then $\mathcal{F}$ is simultaneously diagonalizable.* ◊

2.1.5 Eigenvalue of AB and BA

$p_{AB}(t) = t^{n-m} p_{BA}(t)$ for $A : m \times n$ and $B : n \times m$, $m \leq n$ because:

$$\begin{bmatrix} I_m & -A \\ 0 & I_n \end{bmatrix} \begin{bmatrix} AB & 0 \\ B & 0 \end{bmatrix} \begin{bmatrix} I & A \\ 0 & I \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ B & BA \end{bmatrix} \tag{2.4}$$

2.1.6 Real Similarity

2.5 Lemma. *If $C+iD$ is nonsingular, then there exists $\tau \in \mathbb{R}$ such that $C+\tau D$ is nonsingular.*

2.1.7 Mirsky Theorem

2.6 Theorem. *There exists A has eigenvalues $\lambda_1, \ldots, \lambda_n$ and diagonal $d_1, \ldots, d_n$ iff $\sum \lambda_i = \sum d_i$* ◊

2.2 Chapter 2: Unitary Similarity

***-congruence.** $A \to S^* A S$ where S is not necessarily Unitary.

Unitary. $U^* U = I \iff U$ is Euclidean isometry, $\|x\| = \|Ux\|$ for all $x \iff$ col/rows of U are o.n.b.

2.7 Compactness of $\mathcal{U}$. *$\mathcal{U}$ is the set of all unitary matrices.* ◊

2.8 Selection Principle. *Every sequence of unitary matrices has a convergence subsequence.* ◊

2.9 Lemma. *Rank condition for* $U = \begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix}$

2.10 QR Decomposition. *$A \in M_{m\times n}$, then we have $m \geq n$, $m = n$ and $m < n$.* ◊

Proof. Use Rotation and Householder matrices. □

2.2.1 Unitary Similarity

A is uni.similar to $B \implies \sum_{i,j} |a_{ij}|^2 = \sum |b_{ij}|^2$.

Conversely we have the Word theorem.

2.11 Schur theorem. *Any $A \in M_n$, with eigenvalues $\lambda_1, \ldots, \lambda_n$, let x be a unit eigen-vector for λ_1, then there exists unitary U such that*

$$A = UR_\lambda U^* \tag{2.5}$$

where R is upper-triangular and has $\lambda_1, \ldots, \lambda_n$ on the diagonal. ◊

2.3 Invariant Subspaces and Spectral Theorems

Spectrum. $\sigma(A)$ the set of all eigenvalues of A.

2.12 Proposition (About Matrix Transformation). *For $A = \bar{A}$ that's the complex conjugate of A, $Ax = \bar{A}\bar{x}$.*

For a polynomial $p(A)$, if $\lambda \in \sigma(A)$, then $p(\lambda) \in \sigma(p(A))$. If $\mu \in \sigma(p(A))$, then there exists $\lambda \in \sigma(A)$ such that $p(\lambda) = \mu$.

2.13 Proposition (Normal Matrix). *If $AA^* = A^*A$, then A and A^* share the same eigenvectors, the eigenvalues are λ and $\bar{\lambda}$.*

Proof. If $Ax = \lambda x$, let $B = A - \lambda I$, $x \in N(B)$. $\langle B^*x|B^*x\rangle = \langle x|BB^*x\rangle = \overline{\langle B^*Bx|x\rangle} = \overline{\langle Bx|Bx\rangle} = 0$. □

2.14 Proposition. *Linear Operator on Inf Dim Vector Space might not have eigenvalue. Consider the right-shift operator $S(x_1, \ldots) = S(0, x_1, x_2, \ldots)$.*

2.4 Canonical Forms

2.5 Hermitian Matrix

2.5.1 Characterisation

For complex matrix $A \in \mathbb{M}(\mathbb{C})$, Hermitian matrices are like "real numbers to complex numbers" due to the following decomposition:

2.15 Toeplitz Decomposition. *For any $A \in \mathbb{M}(\mathbb{C})$, there exists unique Hermitian H, K such that $A = H + iK$. It can also be uniquely decomposed as $A = H + S$ where S is skew-Hermitian: $S = -S^*$.* ◊

2.16 Simultaneous Diagonalizable for Hermitian Matrix. *For a commuting class of Hermitian matrix, there exists one unitary U that simultaneously diagonalize $A \in \mathcal{F}$.* ◊

Proof. First, simultaneously make matrices upper-triangular. Then apply the spectral theorem to normal matrices. The Hermitian case is a special case. □

Then it's connected with Positive Definite matrices.

2.5.2 Variational Characterisation

2.17 Courant-Fischer. *Let A be Hermitian and $\lambda_1 \leq \cdots \leq \lambda_n$ be eigenvalues, for $1 \leq k \leq n$, and S a subspace of $\mathbb{C}^n$.*

$$\lambda_k = \min_{S:\dim S=k} \max_{x \in S \setminus \{0\}} \frac{x^* A x}{x^* x} \quad \textit{and} \quad \lambda_k = \max_{S:\dim S=n-k+1} \min_{x \in S \setminus \{0\}} \frac{x^* A x}{x^* x}$$

◊

Proof. Uses Rayleigh's Characterization and subspace intersection lemma. □

2.5.3 Eigenvalue Inequality for Hermitian Matrices

2.18 Weyl's Theorem. ◊

2.19 Cuchy's Interlacing Theorem; Separation Theorem. ◊

2.5.4 Complex Symmetric Matrices

2.6 Norms of Matrices